

Name: Key

3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

rotate counterclockwise by $\frac{\pi}{2}$ and then project to the x-axis

first second

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

or:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

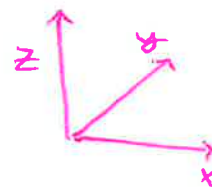
$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

b) Find the 3×3 matrix for the following linear transformation using the standard basis for $\mathbb{R}^3$.

project to the yz-plane

	x	y	z
	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$		
	$\uparrow$	$\nwarrow$	$\nwarrow$
	maps to $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$	stay fixed (those lie in the y-z plane)	



or:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

project to the x -axis and then rotate counterclockwise by $\frac{\pi}{2}$

first second

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \boxed{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}$$

or:

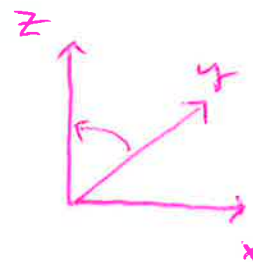
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightsquigarrow \boxed{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}$$

b) Find the 3×3 matrix for the following linear transformation using the standard basis for $\mathbb{R}^3$.

rotate about the x -axis by $\pi/2$ counterclockwise

$$\begin{matrix} & x & y & z \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{0 & -1} \\ 0 & \boxed{1 & 0} \end{pmatrix} \end{matrix}$$



stays fixed

notice that this block is like the 2×2 rotation in part (a).

or:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

$$\boxed{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}}$$